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Differential Calculus and Coordinate Geometry

Implicit Differentiation

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① $x^2 - 4xy + y^2 - 4 = 0 \quad \therefore f(x,y) = 0$

or, $\frac{d}{dx}x^2 - \frac{d}{dx}4xy + \frac{d}{dx}y^2 - \frac{d}{dx}4 = 0$

or, $2x - 4y + 4x\frac{dy}{dx} + 2y \cdot \frac{dy}{dx} - 0 = 0$

or, $4x\frac{dy}{dx} + 2y\frac{dy}{dx} = -2x + 4y$ or, $2x\frac{dy}{dx} + y\frac{dy}{dx} = -x + 2y$

or, $\frac{dy}{dx}(2x+y) = 2y - x$

or, $\frac{dy}{dx} = \frac{2y-x}{2x-y}$

② $x^4 + x^2y^2 + y^3 = 5 \quad \therefore f(x,y) = 0$

$\therefore \frac{d}{dx}x^4 + \frac{d}{dx}x^2y^2 + \frac{d}{dx}y^3 - \frac{d}{dx}5 = 0$

or, $4x^3 + x^2 2y\frac{dy}{dx} + y^2 2x + 3y^2\frac{dy}{dx} - 0 = 0$

or, $x^2 2y\frac{dy}{dx} + 3y^2\frac{dy}{dx} = -4x^3 - y^2 2x$

or, $\frac{dy}{dx}(2x^2y + 3y^2) = -4x^3 - y^2 2x$

or, $\frac{dy}{dx} = \frac{-4x^3 - 2xy^2}{2x^2y + 3y^2}$

(3) $y\cos x = x^2+y^2$ or, $\frac{d}{dx} y\cos x - \frac{d}{dx} x^2 - \frac{d}{dx} y^2 =$

or, $-y\sin x + \cos x \frac{dy}{dx} - 2x - 2y\frac{dy}{dx} = 0$

or, $\frac{dy}{dx}(\cos x - 2y) = 2x + y\sin x$

or, $\frac{dy}{dx} = \frac{2x+y\sin x}{\cos x - 2y}$

(2) $xe^y = x-y$ or, $\frac{d}{dx} xe^y - \frac{d}{dx} x - \frac{d}{dx} y = 0$

or, $e^y + xe^y \frac{dy}{dx} - 1 - \frac{dy}{dx} = 0$

or, $\frac{dy}{dx}(xe^y - 1) = 1 - e^y$ or $\frac{dy}{dx} = \frac{1-e^y}{xe^y-1}$

(4) $2x^2+xy-y^2 = 2$ or, $\frac{d}{dx} 2x^2 + \frac{d}{dx} xy - \frac{dy^2}{dx} - \frac{dz}{dx} = 0$

or, $4x + x\frac{dy}{dx} + y - 2y\frac{dy}{dx} = 0$

or, $\frac{dy}{dx}(x-2y) = -y-4x$ or $\frac{dy}{dx} = \frac{-y-4x}{x-2y}$

(5) $x^3 - xy^2 + y^3 = 1$

or, $\frac{d}{dx} x^3 - \frac{d}{dx} xy^2 + \frac{d}{dx} y^3 - \frac{d}{dx} 1 = 0$

or, $3x^2 - x2y\cdot\frac{dy}{dx} + \frac{dy}{dx}\cdot 3y^2 - 0 = 0$

or, $\frac{dy}{dx}(2xy + 3y^2) = -3x^2$

or, $\frac{dy}{dx} = \frac{-3x^2}{2xy+3y^2}$

⑥ $\frac{x^2}{x+y} = y^2+1$ or, $x^2 = xy^2+x+y^3+y$

or, $\frac{d}{dx}x^2 - \frac{d}{dx}x = \frac{d}{dx}xy^2 + \frac{d}{dx}y^3 + \frac{d}{dx}y$

or, $\frac{d}{dx}\cdot x2y\cdot\frac{dy}{dx} + 3y^2\frac{dy}{dx} + \frac{d}{dx}y = 2x-1$

or, $\frac{d}{dx}(2xy+3y+y) = 2x-1$

or, $\frac{d}{dx} = \frac{2x-1}{2xy+3y+y}$

⑧ $\cos(xy) = 1+\sin y$ or, $\frac{d}{dx}\cdot\cos(xy) = \frac{d1}{dx} + \frac{d\sin y}{dx}$

or, $-\sin xy\cdot x\frac{dy}{dx} + y = 0 + \cos y$

or, $\frac{dy}{dx} = \frac{\cos y - y}{x\sin xy}$

⑨ $\sqrt{x+y} = x^4+y^4$ or, $x+y = x^8+2x^4y^4+y^8$

or, $\frac{d}{dx}x + \frac{d}{dx}y = \frac{d}{dx}x^8 + \frac{d}{dx}2x^4y^4 + \frac{d}{dx}y^8$

or, $1+\frac{dy}{dx} = 8x^7 + 8x^3y^4 + 4y^3\times 8x^3 + 8y^7\frac{dy}{dx}$

or, $\frac{dy}{dx}(1-8y^7) = \frac{8x^7+8x^3y^4+32x^3y^3+8y^7}{}$

or, $\frac{dy}{dx} = \frac{8x^7+8x^3y^4+32x^3y^3+8y^7}{1-8y^7}$

⑩ $e^y\sin x = x+xy$ or, $\frac{d}{dx}e^y\sin x = \frac{dx}{dx} + \frac{dxy}{dx}$

or, $e^y\cdot\frac{dy}{dx}\cdot\sin x + \cos x e^y = 1 + y\frac{dx}{dx} + x\frac{dy}{dx}$

or, $\frac{dy}{dx}(1+ye^y\sin x) = \cos x e^y$

or, $\frac{dy}{dx} = \frac{\cos x e^y}{1+y-e^y\sin x}$

(11) $5y^2+\sin y = x^2$ or, $\frac{d}{dx}5y^2+\frac{d}{dx}\sin y = \frac{d}{dx}x^2$

or, $10y\frac{dy}{dx} - \cos y\frac{dy}{dx} - 2x = 0$

or, $\frac{d}{dx}(10y-\cos y) = 2x$

or, $\frac{dy}{dx} = \frac{2x}{10y-\cos y}$

(12) $x^3+y^3 = 3xy$ or, $\frac{d}{dx}x^3+\frac{d}{dx}y^3-\frac{d}{dx}(3xy)=0$

or, $3x^2+3y^2\frac{dy}{dx} - 3x\cdot\frac{d}{dx} - 3y = 0$

or, $\frac{dy}{dx}(3y^2-3x) = 3y-3x^2$

or, $\frac{dy}{dx} = \frac{3y-3x^2}{3y^2-3x}$

(13) $\sin(x^2y^2) = x^2$ or, $\frac{d}{dx}\sin(x^2y^2) = \frac{d}{dx}x^2$

or, $\cos x^2y^2.(x^2 2y\frac{dy}{dx} + y^2 2x) = 2x$

or, $\frac{dy}{dx} = \frac{2x}{2\cos x^3y^4 + 2\cos x^4y^3}$

(17) $\frac{1}{\sqrt{x}}+\frac{1}{\sqrt{y}} = 1$ or, $\frac{d}{dx}\frac{1}{\sqrt{x}}+\frac{d}{dx}\frac{1}{\sqrt{y}} = 1$

or, $\frac{d}{dx}x^{-\frac{1}{2}}+\frac{d}{dx}y^{-\frac{1}{2}} = \frac{d}{dx}1$

or, $-\frac{1}{2}x^{-\frac{3}{2}} - \frac{1}{2}y^{-\frac{3}{2}}\frac{dy}{dx} = 0$

or, $\frac{dy}{dx} = \frac{x^{-\frac{3}{2}}}{-y^{-\frac{3}{2}}}$

Parametric Differentiation

① $x = t^2+1$, $y = t^2+t$ $\therefore \frac{dx}{dt} = 2t$ $\therefore \frac{dy}{dt} = 2t+1$

$\therefore \frac{dy}{dx} = \frac{2t+1}{2t}$

② $x = t^3+1$, $y = t^2-t$ $\therefore \frac{dx}{dt} = 3t+1$ $\therefore \frac{dy}{dt} = 2t-1$

$\therefore \frac{dy}{dx} = \frac{2t-1}{3t+1}$

③ $x = e^t$, $y = te^{-t}$ $\therefore \frac{dx}{dt} = e^t$ $\therefore \frac{dy}{dt} = -te^{-t} + e^{-t}$

$\frac{dy}{dx} = \frac{e^{-t} - te^{-t}}{e^t}$

④ $x = t^2+1$, $y = e^t - 1$ $\therefore \frac{dx}{dt} = 2t$ $\therefore \frac{dy}{dt} = e^t$

$\frac{dy}{dx} = \frac{e^t}{2t}$

⑤ $x = t - \ln t$, $y = t + \ln t$ $\therefore \frac{dx}{dt} = 1 - \frac{1}{t}$ $\therefore \frac{dy}{dt} = 1 + \frac{1}{t}$

$\frac{dy}{dx} = \frac{1+\frac{1}{t}}{1-\frac{1}{t}} = \frac{t+1}{t-1}$

⑥ $x = \cos t$, $y = \sin 2t$ $\frac{dx}{dt} = -\sin t$ $\therefore \frac{dy}{dt} = \cos 2t$

$\frac{dy}{dx} = \frac{\cos 2t}{-\sin t}$

⑦ $x = \sqrt{t}$, $y = 2t+7$ $\frac{dx}{dt} = \frac{1}{2\sqrt{t}}$, $\therefore \frac{dy}{dt} = 2$

$\therefore \frac{dy}{dx} = \frac{2}{\frac{1}{2\sqrt{t}}} = \frac{1}{\sqrt{t}}$

(8) ~~sect, y=~~ $x = \sec t$, $y = \tan t$ $\therefore \frac{dx}{dt} = \sec t \tan t$

$\therefore \frac{dy}{dt} = \sec^2 t$ $\therefore \frac{dy}{dx} = \frac{\sec^2 t}{\sec t \tan t} = \frac{\sec t}{\tan t}$

$= \frac{\frac{1}{\cos t}}{\frac{\sin t}{\cos t}} = \frac{1}{\sin t} = \operatorname{cosec} t$

(9) $x = t + \cos t$, $y = 1 + \sin t$ $\therefore \frac{dx}{dt} = 1 - \sin t$

$\frac{dy}{dt} = \cos t$ $\therefore \frac{dy}{dx} = \frac{\cos t}{1 - \sin t}$

(10) $x = \sinh t$, $y = \cosh t$ $\therefore \frac{dx}{dt} = \cosh t$

$\therefore \frac{dy}{dt} = \sinh t$ $\therefore \frac{dy}{dx} = \frac{\sinh t}{\cosh t} = \tanh t$

(11) $x = \frac{1}{2}t^2 + 1$, $y = \frac{1}{3}t^3 - t$ $\therefore \frac{dx}{dt} = t + t^2$

$\therefore \frac{dy}{dt} = t + t^3$ $\therefore \frac{dy}{dx} = \frac{1 + t^2}{1 + t}$